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The Oxidation Behavior of Several Ti–Al Alloys at 900°C in Air

M. Schmitz-Niederau* and M. Schütze†

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Ti–23Al, Ti–50Al and Ti–50Al–2Nb (at.%) were oxidized in air at 900°C for times up to 1130 hr. The resulting oxide scale structures were analyzed in great detail by metallographic and microprobe investigations and the Al_2O_3 structures in the complex oxide scales were correlated with the course of the thermogravimetric curves. It appears that in order to achieve long-term protective behavior of the scales, it is necessary to stimulate the formation of a thin Al_2O_3 barrier at the scale–metal interface and not at a position in the outer part of the scale. The Nb effect seems to be mostly due to this stimulation of an Al_2O_3 layer at the interface.

KEY WORDS: Ti–Al alloys; scale structures; protective effect.

INTRODUCTION

There is an increasing interest in Ti–Al alloys from a technical point of view because of their light weight and their remarkable high-temperature strength, suggesting their use in gas turbines and as parts in automotive engines. Consequently, there has been a large number of investigations into their mechanical properties and aspects of production. Although their use at elevated temperatures is particularly recommended there are also a number of investigations of the oxidation behavior, which are summarized in Refs. 1–3. It was shown that oxidation resistance in air usually is sufficient for temperatures up to 700°C. When adding Nb or certain other elements like, for example, W or Mo, the oxidation resistance can be improved

*Lurgi Ol Gas Chemie GmbH, D-60295 Frankfurt/Main, Germany.
†Karl-Winnacker-Institut der DECHEMA e.V., D-60061 Frankfurt/Main, Germany.

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Table I. Composition of the Alloys Investigated (at.%)

	Al	Ti	Nb	Fe	C	O
Ti–50Al	50.7	49.2	0.004	0.007	0.019	0.12
Ti–50Al–2Nb	50.2	47.5	1.99	0.016	0.026	0.24
Ti–23Al	23	77	—	—	—	—

significantly[4] allowing use at temperatures up to about 800°C. Another method for improving the oxidation resistance is the use of the chlorine effect.[5,6] However, there exists considerable scatter in the oxidation data in the literature and much debate on which type of oxide is formed on Ti–Al alloys. In particular, there is some uncertainty concerning the long-term behavior of these alloys.

The present work deals with three typical titanium-aluminide alloys, which may be taken as characteristic for certain groups among the range of these materials. The first is an α_2-Ti_3Al alloy with 23 at.% Al. The second is stoichiometric Ti–Al with 50 at.% Al, and the third is a Ti–Al alloy with 50 at.% Al and 2 at.% Nb. According to the literature, no protective oxide scale would be expected at 900°C in air for the α_2-alloy, while good protection should be present for the Nb-containing version. Stoichiometric Ti–Al may show slow oxidation rates at the beginning, but accelerate oxidation after reaching a breakaway point.[7]

The aim of this work was to investigate the oxidation kinetics of these three materials at 900°C in air, which is regarded, at present, as a temperature marking the upper limit of potential practical use of titanium-aluminide alloys. Special emphasis was placed on a detailed examination of the oxide scale structures formed in order to understand the type of kinetics observed under these conditions and of the role of the scale structure for the oxidation rate.

EXPERIMENTAL PROCEDURES

The composition of the alloys investigated is given in Table I. The alloys were in the as-cast condition and the microstructures of Ti–50Al and Ti–50Al–2Nb are shown in Fig. 1. In both cases, lamellar colonies consisting of γ-Ti–Al and α_2-Ti_3Al were embedded in a dendritically solidified γ-matrix. The microstructure of the Ti–23Al alloy also showed a lamellar structure with α_2 dominating. For the oxidation tests, cube-shaped specimens (edge length, 5 mm) were produced with a grit finish of 4000. Thermogravimetric tests were performed in a Sartorius micro M25 D-V microbalance at 900°C in laboratory air. The heating rate was 10°C/min and cooling occurred in the furnace after switching off. Marker experiments

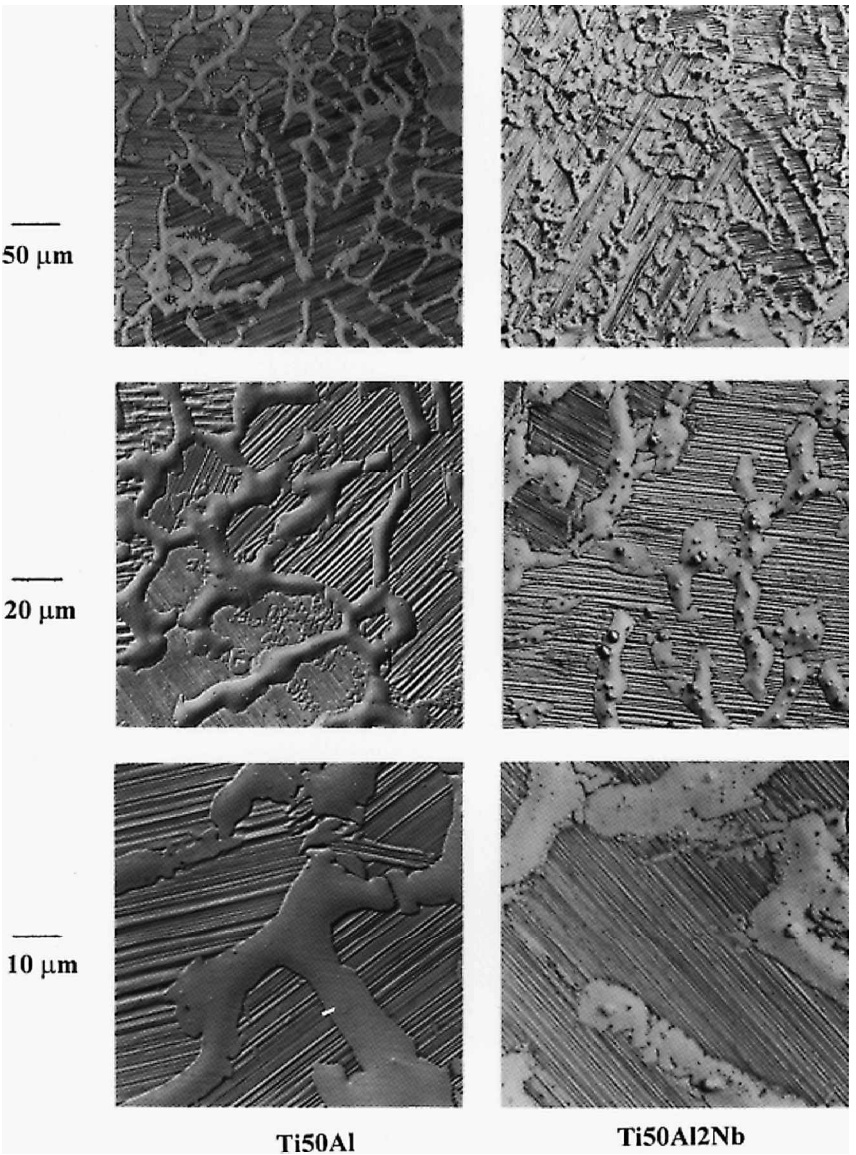


Fig. 1. Microstructures of Ti–50Al and Ti–50Al–2Nb in the present investigations.

were performed with some of the specimens in order to determine the oxide growth direction. For these experiments the specimens received a thin porous Pt coating by cathodic deposition from a solution of 5 g $H_2(PtCl_6) \cdot 6H_2O$ in 160 ml water using a Pt anode and a current density of $5\ mA/cm^2$.

Much emphasis was placed on a detailed examination of the scale microstructures. For these investigations, after the oxidation tests, the specimens were first covered with a thin gold coating by sputtering, which provided electrical conductivity for subsequent galvanic coating with Ni.[8] The latter procedure significantly improves the detectability of the optical features in surface scales. The specimens were embedded in a conducting mount and polished down to $1/4\ \mu m$. The metallographic sections were then coated with a thin interference layer of ZnSe for use with interference layer metallography in the detection of different phases of the microstructure. In order to obtain information about the composition of the different phases encountered in the oxide scales and in the metal subsurface zones a CAMECA SX50 electron-probe microanalysis (EPMA) was used for analysis. The standards were pure metals and the oxygen content was obtained by subtracting from 100%.

RESULTS

Oxidation Kinetics

The results from the thermogravimetric measurements are plotted in Figs. 2–4. Since it was expected that scale growth occurs by a diffusion-controlled mechanism, a parabolic plot was used. It becomes evident that there is no simple parabolic behavior and that the slope (i.e., the ''parabolic rate constant'') changes during oxidation. For Ti–50Al there are three regions in the curve in which the slope of the parabolic plot remains practically constant. The first region covers the time from 0 to 24 hr, the second region covers the time from 30 to 450 hr, and the third region covers the time from 700 hr until the end of the experiment after about 1000 hr. For all three time regions, parabolic rate constants were determined; they are given in Fig. 2.

For Ti–50Al–2Nb, some five regions were observed with different temporary parabolic rate constants, whose values are given in Fig. 3. The first region ranges from 0 to 30 hr, the second from 30 to 350 hr, the third from 350 to 850 hr, the fourth from 850 to 1000 hr, and the fifth until the end of the test at 1130 hr. Region 2 is comparable to region 2 of Ti–50Al. All other values are lower for Ti–50Al–2Nb. It is interesting to note, however, that even for the Nb-containing alloy, a kind of a temporary breakaway effect occurs after about 850 hr lasting until about 1000 hr (region 4).

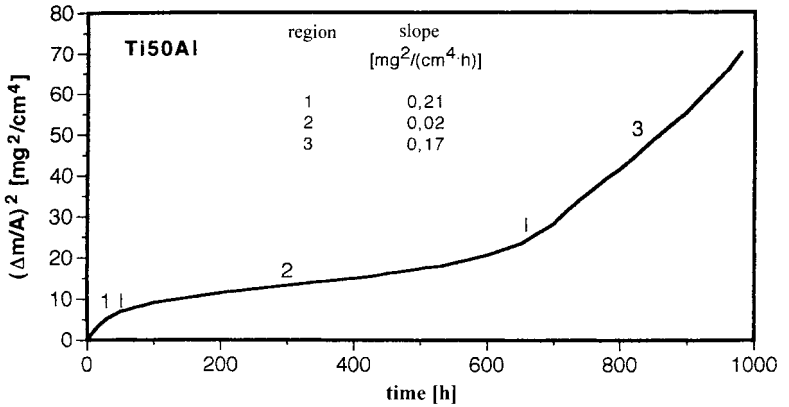


Fig. 2. Parabolic plot of mass gain vs. time during oxidation of Ti–50Al at 900°C in air. Three characteristic time regions were observed; for each region a mean slope was determined.

In addition, for Ti_3Al there are different time regions with different slopes in the mass-gain curve (Fig. 4). Again, the rate constants are provided in the figure. The first time region lasts until 8 hr, the second until 30 hr, the third from 31 to 49 hr, and the fourth from 50 hr until the end of the test after 110 hr.

As a common feature of the oxidation curves of all three materials, phase 1 shows the highest oxidation rates. Similar rates are observed in the breakaway region (region 3 for Ti–50Al and region 4 for Ti–50Al–2Nb).

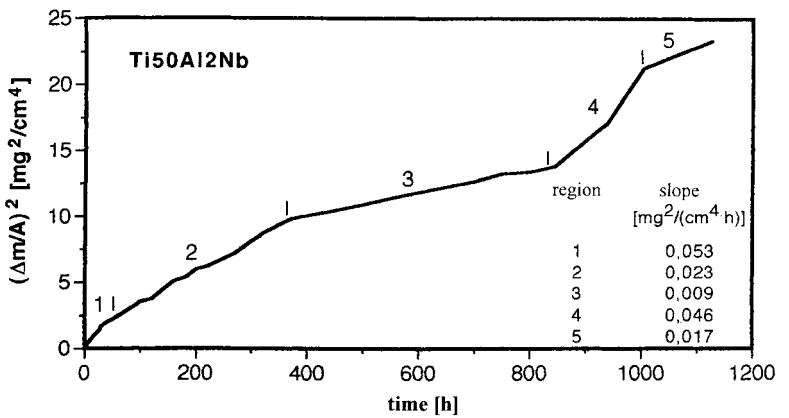


Fig. 3. Parabolic plot of mass gain vs. time for Ti–50Al–2Nb. Five characteristic time regions were assigned to the thermogravimetric curve.

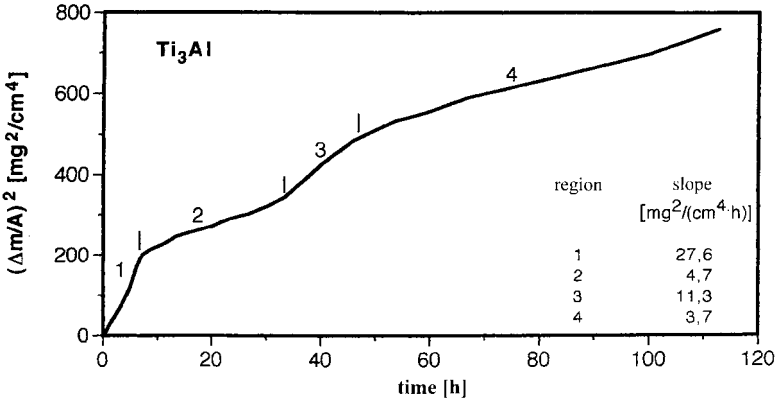


Fig. 4. Parabolic plot of mass gain vs. time for Ti–23Al.

The highest rates are found for Ti–23Al, which is not surprising due to the lower Al content. These rates are at least about two orders of magnitude higher than in the breakaway periods of the other materials so that no protective effect of the oxide scale is present. While the breakaway period lasts for a longer time in the case of Ti–50Al, with no indication of a change back to region-2 behavior, there is a period of only 150 hr for Ti–50Al–2Nb where the oxidation rate takes higher values (region 4). After this period, the oxidation rate decreases again.

It may be argued that if the oxidation rate cannot be described by a single parabolic rate constant during the whole test, it is not permissible to calculate a parabolic rate constant. As former investigations show,[1,7] it is diffusion in the oxide that determines the oxidation rates. There may, however, be changes in the oxide structure and composition during oxidation time leading to changes in the scale growth rates, a fact that will be discussed in more detail later. Thus, the introduction of temporary parabolic rate constants seems permissible.

Scale Structure

For the characterization of the scale structures, a larger number of micrographs were investigated by metallography and microprobe; the results were evaluated and plotted in the form of schematic representations. Examples of the appearance of the scales on Ti–50Al after breakaway (region 3) are given in Fig. 5 (the dark gray areas are alumina and the bright gray areas titania). The bright field in the lower part of the photographs is the metal. These photographs reveal the main features of the oxide scales

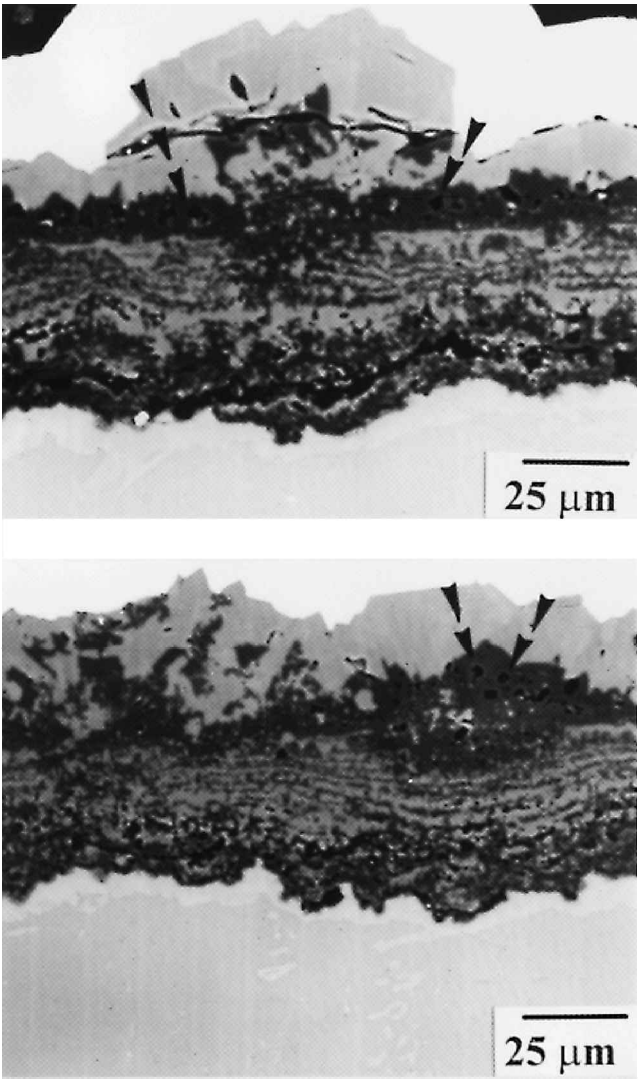


Fig. 5. Metallographic section of the oxide scale on Ti–50Al after breakaway (960 hr in air at 900°C). TiO_2 appears light gray; Al_2O_3 appears dark. The metal–subsurface zone contains a bright seam. Pores in the scale are shown by arrows.

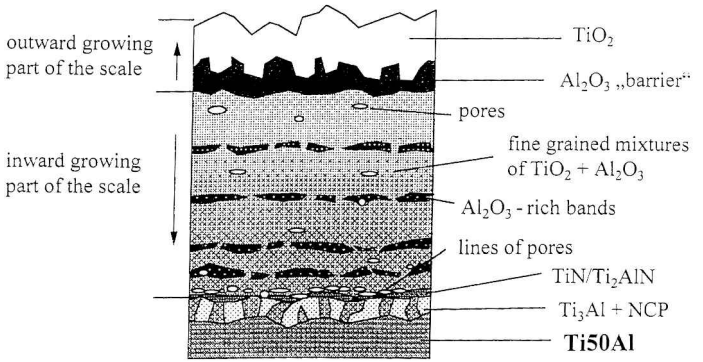


Fig. 6. Schematic representation of the scale structure on Ti–50Al after oxidation in air at 900°C before breakaway.

on Ti–50Al after longer oxidation times. The outermost partial layer is more or less pure TiO_2, followed by a band of Al_2O_3, and then by a layered structure with Al_2O_3 enrichment in a fine TiO_2–Al_2O_3 mixture. Further down in the scales, the number of Al_2O_3 phases dominates and exceeds the amount of TiO_2. However, with the exception of the rather thick outer Al_2O_3 band, before breakaway there is no continuous Al_2O_3 partial layer in the inner part of the scales. As shown in Ref. 7, the alumina band is continuous before breakaway occurs (i.e., in region 2) and is interrupted after breakaway (i.e., in region 3). The same observations were made in the present work; Fig. 5 shows these locally interrupted Al_2O_3 bands. At sites of these local interruptions, Al_2O_3 particles are found in the outer, nominally pure TiO_2 layer, together with a locally increased scale thickness. Furthermore, pores are observed in the scales, which is important for discussions of scale adhesion and cracking. Their mechanism of formation is discussed elsewhere.[9] The scale structure, which is found before breakaway, is similar to that shown in Fig. 5, with the difference, however, that the Al_2O_3 band is closed and that no local TiO_2 nodules grow out of the scales. The schematic representation of the structure is given in Fig. 6. This figure also contains some more information from microprobe investigations, indicating that a discontinuous "layer" of nitrides is present at the scale–metal interface. Metallographic investigations of the metal–subsurface zone show that oxidation leads to changes in that a two-phase layer is formed in the metal (Fig. 7). Microprobe results reveal that the two phases are α_2, containing a large amount of dissolved oxygen and a new cubic phase (NCP) often also called the X- or Z-phase, which was recently described. This information is summarized in Fig. 6.

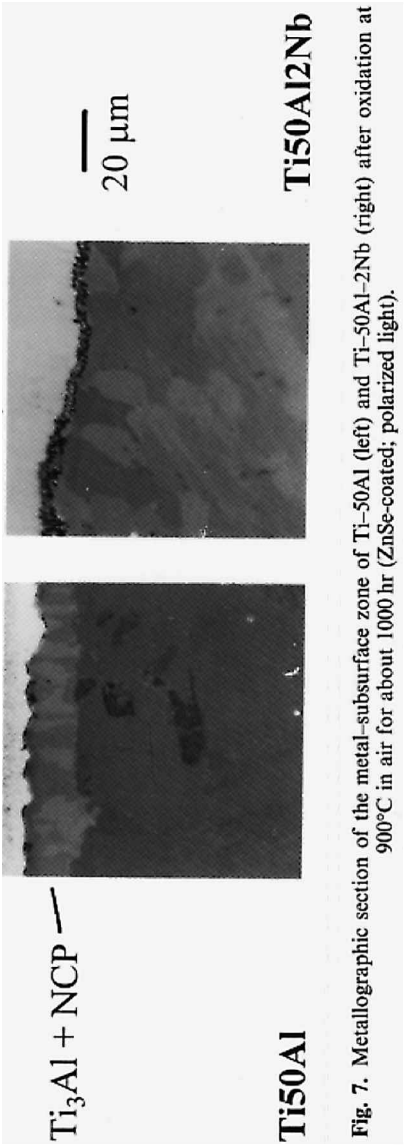


Fig. 7. Metallographic section of the metal–subsurface zone of Ti–50Al (left) and Ti–50Al–2Nb (right) after oxidation at 900°C in air for about 1000 hr (ZnSe-coated; polarized light).

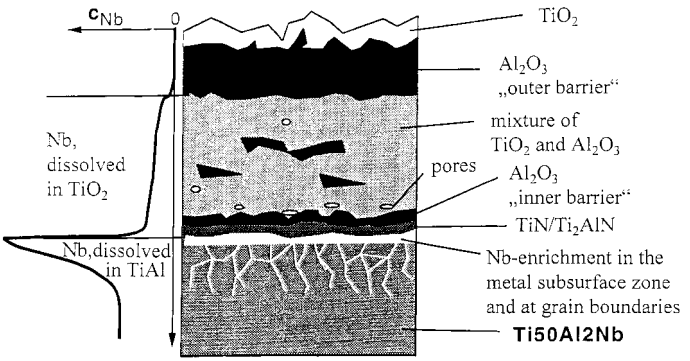


Fig. 8. Schematic representation of the scale structure on Ti–50Al–2Nb after oxidation in air at 900°C and after longer oxidation times. The Nb concentration is shown as a qualitative curve.

The scale structure is quite different for Ti–50Al–2Nb. Schematics are given in Figs. 8 and 9, which show that, contrary to Ti–50Al, there is an inner Al_2O_3 barrier after longer times close to the scale–metal interface above a discontinuous nitride layer. Often this Al_2O_3 barrier was only very thin (about 1 μm) and could sometimes only be detected by microprobe line scans. For this alloy, the Nb distribution in the scale structure and the metal subsurface zone was of special interest. The microprobe results are given in Table II. These also reveal that there is Nb enrichment at grain boundaries in the metal–subsurface zone and in a metal zone adjacent to the scale–metal interface. The Nb concentrations at these sites can reach double the values of the original material. The formation of new phases in the metal–subsurface zone was not observed (cf. also Fig. 7). The Nb content, as well as the Al content, in TiO_2 dropped from the inside of the scale in the outer

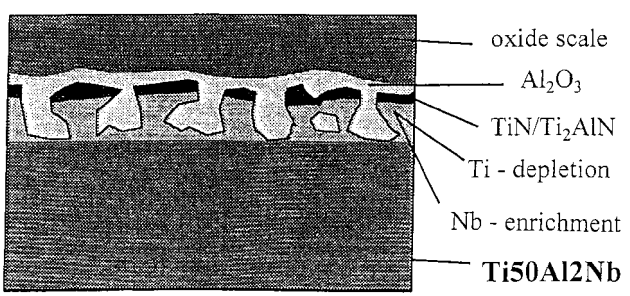


Fig. 9. Structure of the metal–subsurface zone after longer oxidation times.

Table II. Results from the Microanalytical Investigations (at.%) of Different Parts in the Scale–Metal Structure on Ti–50Al–2Nb After 1000 hr at 900°C in Air

Position	Nb	Ti	Al	O (=Rest)
Al_2O_3	0.00	2.02	38.93	59.05
TiO_2				
Outer part	0.58	33.35	0.49	65.58
Inner part	0.96	32.17	1.18	65.70
Metal subsurface zone	4.40	54.89	28.85	11.86
Grain interior	2.38	47.73	48.19	1.71

direction. The crystals in the top layer are smaller than in the case of Ti–50Al.

The structure of the scale on Ti–23Al and of the metal–subsurface zone are given in Fig. 10. In Fig. 11 the structures are again summarized in a schematic. The structure is somewhat similar to that of Ti–50Al, but there is no continuous Al_2O_3 barrier, and, in the TiO_2–Al_2O_3 mixture beneath, a certain outer Al_2O_3 enrichment of TiO_2 dominates. At the scale–metal interface, there is a TiN layer followed by a layer, which, from its composition, could be described by $(Ti_{0.75}Al_{0.25})_3O_2$. The metal–subsurface zone consists of Ti_3Al with a large amount of dissolved oxygen.

In the marker experiments, the growth directions of the partial layers were determined. It turned out that the outer pure TiO_2 layer grows in an outward direction, while the remaining, more complex, structures containing Al_2O_3 grow in an inward direction. The proportion between inward and outward growth is 3 to 1 for Ti–50Al and Ti–50Al–Nb and 2 to 1 for Ti–23Al, as long as breakaway had not yet occurred. After breakaway of the scale on Ti–50Al, the situation becomes irregular, with local increases in scale thickness (nodule formation) where TiO_2 growth dominates and where large Al_2O_3 particles are embedded in the outer, thick TiO_2 nodules. With increasing oxidation time, these breakaway nodules increase in number and size and become connected to form a thicker scale with irregular thickness values. A ratio between the outward and inward thickness growth can no longer be exactly defined.

DISCUSSION

The kinetics curves of the three different alloys are plotted together in Fig. 12. This figure also contains the curves typical for pure TiO_2 scales (i.e., scales on titanium[12]) and for an alumina former (e.g., scale on a high-aluminum-containing, Ni-base alloy[12]). It becomes evident that Ti–23Al is close to "pure TiO_2 kinetics," which is not surprising from the features of

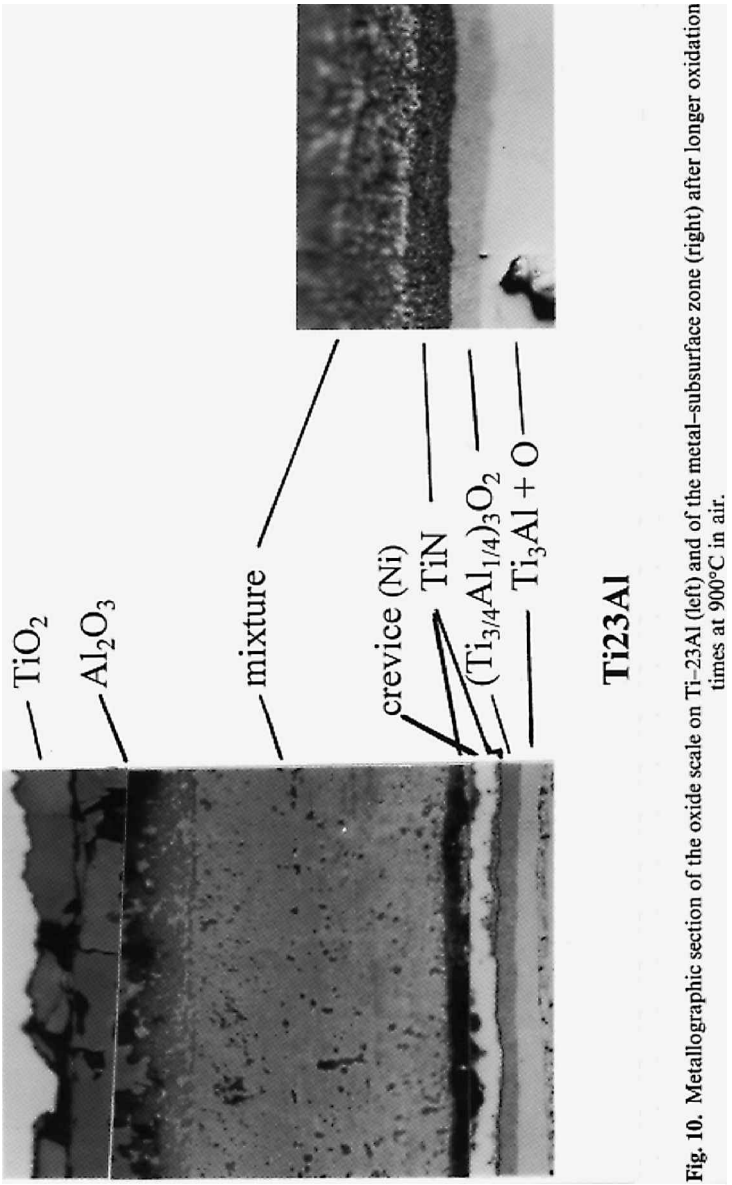


Fig. 10. Metallographic section of the oxide scale on Ti–23Al (left) and of the metal–subsurface zone (right) after longer oxidation times at 900°C in air.

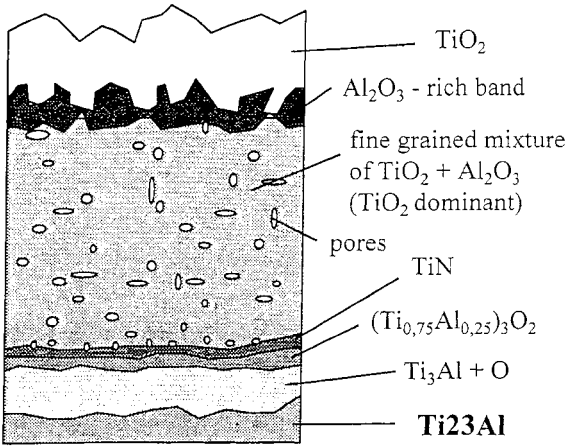


Fig. 11. Schematic representation of the scale structure on Ti–23Al after 110 hr oxidation at 900°C in air.

the scale structure. The alumina-rich band (cf. Figs. 10 and 11) is not continuous, so that TiO_2 paths exist through this band. An influence of alumina as a barrier in this scale can exist only in that the diffusional cross section for rapid transport in the TiO_2 phase is narrowed by the presence of alumina particles of this discontinuous band. The effect of these particles seems, however, to be rather marginal. The aluminum particle distribution density in the inward-growing scale evidently is not sufficient to provide any significant diffusion barrier. Thus, Ti–23Al stands for a group of Ti–Al alloys

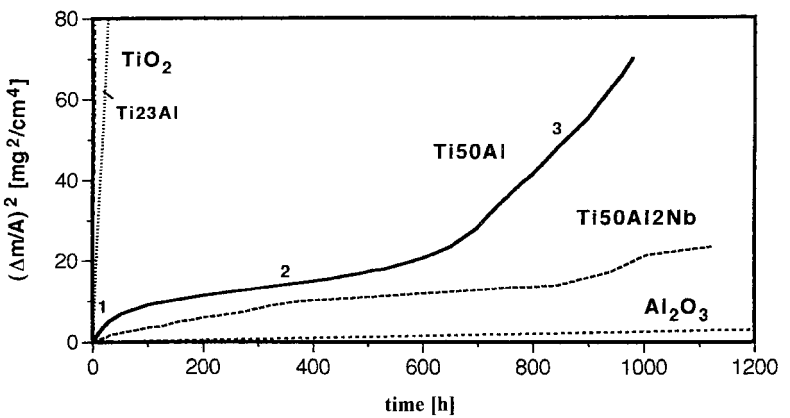


Fig. 12. Parabolic plot of mass gain vs. time for all three materials. Furthermore, the slopes for TiO_2 and Al_2O_3 formers are indicated.

for which the presence of even 23 at.% Al does not provide protection. Nevertheless, as Fig. 4 shows, the diffusional cross section between the alumina particles seems to vary with time, leading to changes in the slope and a certain positive effect is established after about 9 hr (region 2 and following regions). In region 1 (i.e., up to 9 hr) the oxidation rate is very close to pure TiO_2 kinetics.

For Ti–50Al–2Nb in region 3, the slope is practically identical with the kinetics of a genuine alumina former (cf. Figs. 3 and 12). The beginning of region 3 represents about the time when a significant alumina barrier is established close to the scale–metal interface and when optimum protection can be expected. In regions 1 and 2, oxidation rates are also low, but evidently some faster diffusion paths through the alumina partial layers, in the form of TiO_2 channels, must exist.

It is somewhat surprising that even for this alloy, a kind of temporary breakaway effect is present. This effect is not really dramatic, but as Fig. 3 shows, it is significant. At present, it can only be speculated about its origin, but it seems likely that growth stresses in the scales build up, which may lead to the formation of cracks in the scales accelerating oxidation until the original scale structure has been restored by healing.

There has been some debate on how Nb improves the oxidation resistance of Ti–Al alloys. It was suggested that Nb^{5+} ions in the TiO_2 lattice decrease the number of oxygen vacancies and titanium interstitials, which are both regarded as responsible for the high growth rates of TiO_2 scales.[12] Another explanation could be an increase in the aluminum activity in the metal–subsurface zone because of the presence of Nb stimulating the formation of an Al_2O_3 barrier at the scale–metal interface. It is also conceivable that the solubility of Al cations in TiO_2 is decreased by Nb^{5+} doping, thus stimulating the precipitation of an Al_2O_3 barrier in the inner part of the scale of at the scale–metal interface by a similar mechanism as described for the intermediate-barrier formation in Ref. 7. The formation of the intermediate alumina barrier on Ti–50Al has been explained by the precipitation of alumina in a layer where the solubility limit was exceeded by the Al concentration in TiO_2. The solubility limit decreases with increasing oxygen partial pressure[13] or decreasing number of oxygen vacancies. In other words, if the oxygen-vacancy concentration is decreased by Nb doping, the layer in which Al_2O_3 precipitation occurs may be shifted to the inner parts of the scale, and in the extreme, to the scale–metal interface. The present results do not allow a definite conclusion on these points. All of the points can help explain the effect, since considerable Nb enrichment was found in the metal–subsurface zone and in addition, Nb was present in significant amounts in the inner part of the oxide scale. The most likely positive effect of Nb is the stimulation of the formation of an inner alumina barrier by decreasing the Al solubility in TiO_2. Decreasing the diffusion rates in TiO_2

is possibly not the primary effect. Recent results of Al activity measurements indicate that the subsurface zone activity of Al is rather decreased by the presence of Nb,[14] so that this effect cannot contribute to the explanation of the Al_2O_3 barrier. Only in region 3 is true alumina kinetics reached and it seems to be necessary that Nb becomes enriched in the metal–subsurface zone to a certain extent or must become incorporated in the oxide before stimulation of an alumina barrier formation occurs.

The situation for the scales on Ti–50Al has already been discussed earlier.[7] As a main feature, a temporary alumina barrier is formed after an incubation time between the outward- and the inward-growing parts of the scales. This barrier becomes dissolved in a later period as the oxygen partial pressure gradient in the scale changes with increasing thickness, leading to a local increase of Al solubility in TiO_2, which is dependent on the oxygen partial pressure (see above). The result is the observed breakaway effect.

The three alloys investigated represent three different groups of Ti–Al alloys. Ti–23Al represents those alloys that never form a continuous, protective alumina barrier. Ti–50Al represents those alloys forming a temporary alumina barrier, which loses its protectiveness by dissolution in a later period. Ti–50Al–2Nb represents those alloys where a long term Al_2O_3 barrier is formed after a certain incubation period. As a main feature, this barrier is situated in the scale at a position close to the scale–metal equilibrium pressure. This inner partial layer, thus, is not endangered by dissolution and possesses long-term stability, unless it is cracked by growth or external scale stresses. This fact may be the key to long-term oxidation protection of Ti–Al alloys. Thus, all measures stabilizing a continuous Al_2O_3 partial layer at the scale–metal interface will be successful in improving oxidation resistance of Ti–Al alloys for service conditions, as this partial layer is immune to dissolution, since the p_{O_2} at this site always remains close to the Al_2O_3 equilibrium partial pressure. Besides alloying, special surface treatments may achieve this aim, among which, in particular, is Cl enrichment in the metal surface by a combined PVD/annealing treatment or by ion implantation; both have proved to yield a significant improvement in oxidation resistance.[6]

In the investigations, nitrides were encountered also at the scale–metal interface. It was sometimes speculated that even these nitrides may have a positive effect in the protection of Ti–Al. As they usually do not, however, form a continuous layer and as they may become converted to TiO_2 with increasing oxygen partial pressure when the scale grows inwards,[9] it is not believed that protection results from these nitrides.

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